MILITARY SPECIFICATION

FIN ASSEMBLY, M4A1

1. SCOPE

1.1 This specification covers Fin Assembly, M4A1.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS

Federal
Federal Test Method Standard No. 151 - Metals; Test Methods

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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2.1 (Cont'd)

DRAWINGS

Munitions Command

D10523464 - Fin Assembly, M4A1
IEL-10523464 - Index of Inspection Equipment List for Fin Assembly, M4A1

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 First article sample.-Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.3 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, referenced specifications and the following:

3.3.1 Metal defective.-Soundness of the material shall be as required by the applicable material specifications.

3.3.2 Surface finish.-The requirement for surface finish are as detailed on the applicable drawing.

3.3.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings and shall be in accordance with the provisions of MIL-A-2550.

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3.3.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.3.5 Foreign matter.-All parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 15 metal parts assemblies as follows:

10 Fin assemblies, complete
5 Shanks

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

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4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

* 4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105. Lot numbering shall be as specified by MIL-STD-1168.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

* 4.3.2.1 Control tool manufacture.-Dimensions marked "control tool manufacture" shall be gaged at the beginning of production, whenever tooling is changed or as a minimum as defined below.

(a) Location of flash holes - Select and gage one assembly from each machine every two hours.

(b) Size of flash holes - Select and gage one assembly from each machine every two hours.

(c) Fin blade dimensions - Select and gage one assembly, every 5000 pieces, from each die cast operation.

1. All material produced prior to periodic inspection for items (a) and (b) is acceptable; however, corrective action must be taken whenever a defective characteristic is found.

2. All material produced subsequent to periodic inspection for item (c) will be rejected and screened whenever a defective characteristic is found and corrective action shall be taken.

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4.3.2.2 Shank - Drawing C10523458 covering a detail of Drawing D10523464.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Diameter of knurled area	Gage
202. Diameter forward of knurled area	Gage
203. Length from front face to start of diameter forward of knurled area	Gage
204. Length of knurled area, min.	Gage
205. Length of diameter forward of knurled area	Gage
206. Length, total	Gage
207. Radii or chamfers missing or incorrect	Visual (see 6.1)
208. Knurling missing or incorrect	Visual
209. Surface finish, improper	Visual (see 6.1)
210. Burr	Visual
211. Foreign matter	Visual
212. Metal defective	Visual

4.3.2.3 Assembly - Drawing D10523464

CRITICAL	
None defined	
MAJOR	
101. Concentricity of large diameter of fin blades with pitch diameter of front thread, including perpendicularity of front shoulder	Gage
102. Cartridge cavity not functionally concentric with pitch diameter of front thread	Gage
103. Minor diameter of rear thread not functionally concentric with pitch diameter of forward thread	Gage
104. Rear counterbore not functionally concentric with pitch diameter of forward thread	Gage

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
105.	Pitch diameter of external thread, min.	Gage
106.	Major diameter of external thread, min.	Gage
107.	Pitch diameter of internal thread, max.	Gage
108.	Minor diameter of internal thread, max.	Gage
109.	Diameter over fin blades, large	Gage
110.	Diameter of cartridge cavity	Gage
111.	Diameter of recess rear of fin blades	Gage
112.	Depth of cartridge cavity	Gage
113.	Diameter at basic length	Gage
114.	Length of effective external thread, min.	Gage
115.	Length of effective internal thread, min.	Gage
116.	Wall thickness, min.	Gage
117.	Step between fin and shank, max.	Gage
118.	Location of fin blades (angular)	Gage
119.	Ignition flash hole(s) missing or excessive in number	Visual
120.	Threads damaged or not full	Visual
121.	Burr in cartridge cavity	Visual
122.	Metal defective	Visual
123.	Fin blades distorted or damaged	Visual
MINOR		
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of internal thread, min.	Gage
203.	Diameter of external thread pilot	Gage
204.	Diameter of internal thread counterbore	Gage
205.	Diameter of hub	Gage
206.	Depth of internal thread counterbore	Gage
207.	Length of large diameter of fin	Gage
208.	Length from front face to end of external thread	Gage
209.	Length from front face to shoulder	Gage
210.	Length of internal thread, max.	Gage
211.	Length of internal thread, pilot	Gage
212.	Length, total	Gage
213.	Surface finish, improper	Visual (see 6.1)
214.	Radii or chamfers missing or incorrect	Visual (see 6.1)
215.	Marking missing, incorrect or illegible	Visual
216.	Burr (except as otherwise classified)	Visual
217.	Foreign matter	Visual

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4.3.2.4 Packed Carton prior to sealing

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Support(s) missing or damaged	Visual
202. Divider(s) missing or damaged	Visual
203. Improperly packed	Visual

4.3.2.5 Packed Carton after sealing

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Cartons damaged (cut or damaged through all layers)	Visual
203. Carton not properly sealed	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Finished aluminum components.-The sample size and acceptance criteria shall be as specified in the applicable material specification. Failure to comply with the requirements of the applicable specifications shall be cause for rejection of the lot.

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4.3.4 Inspection equipment.-Index of Inspection Equipment List Number 10523464 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation on the applicable IEL. Reference is made to the inspection equipment list index for definition of coding format.

4.4 Test methods and procedures.

* 4.4.1 Mechanical properties.-The method and procedure for performing these tests shall be in accordance with the requirements of Federal Test Method Standard No. 151 and the applicable material specification.

5. PREPARATION FOR DELIVERY

5.1 Packing.

* 5.1.1 Level C.-Fin assemblies shall be packed in accordance with the requirements of MIL-STD-1169.

* 5.2 Labeling and Marking.-Labeling and marking shall be in accordance with the requirements of MIL-STD-1169.

5.3 Data card.-Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

* The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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